Presidential Facebook Ad Spending Pre and Post Debate

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Introduction

The 2024 Presidential Campaign was nothing short of of a Hollywood-caliber drama. While it was easy to focus on the constant stream of eye-popping headlines (incumbent President suddenly dropping out, a nearly successful assassination attempt, and one of the candidates being an active criminal defendant to name a few), some of the more mundane irregularities were not as spotlighted. The debate cycle started irregularly early with the first one in June, and only one debate following the convention. Since the candidates who ended up being on the ballot in November only had one chance to defend their positions in real-time, we wanted to take a look at the effects the debate had on the campaigns as opposed to the voters. We examined ad spending data to see what topics the candidates put their money towards both before and after the debate.

Research Questions

- Did the debate influence how the candidates budgeted their ad buys?
- Which topics seemed to get the most attention before and after the debate?
- Did topic classification by keyword-based classification vary from classification through OpenAI?

Methods

Sample

- We used the Wesleyan Media Project's ad data, specifically the Daily Delta dataset to attain the ad content and spending during the observed time period.

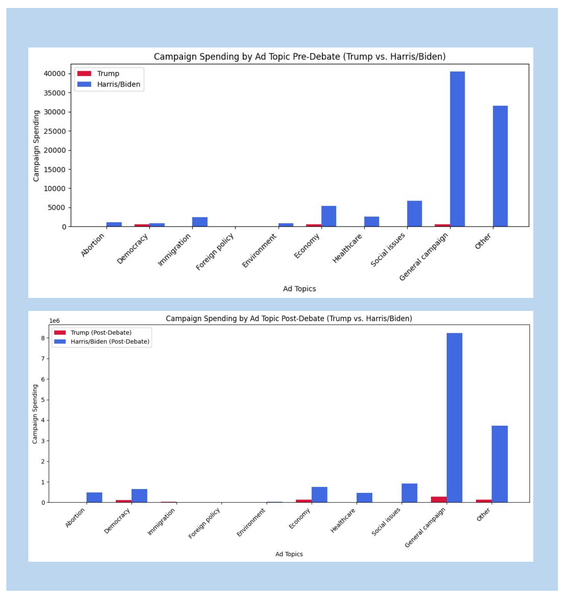
Measures

- We took two approaches to analyzing and categorizing the data by topic: one was identifying hard-coded "key-words" matches in the ad to a corresponding topic, and the other was matching the ad to a topic through ChatGPT version 4o mini
- We eliminated values that did not contain any relevant data on the content of the ad
- In the ChatGPT version, the prompt would be given the content of an ad and then the model would return one or more classifications for the ad, such as "abortion", "healthcare". For the keyword search we only labelled each ad with one classification.
- We calculated the median spending in between the low and high ranges for each ad then calculated the total spending for each category.

Results

- Overall, campaign spending on Facebook ads were much higher for the Harris/Biden campaign than the Trump campaign.
- A large majority of the spending was classified under the "General Campaign" category. Many of the advertisements that fell under this category were simply calling for support or donations.
- Social issues and the economy were other hot button issues that were heavily endorsed in the Harris/Biden ad spending
- While Trump more endorsed advertisements on the economy and democracy.

Figure 1. ChatGPT Based Topic Classifications Pre and Post Debate

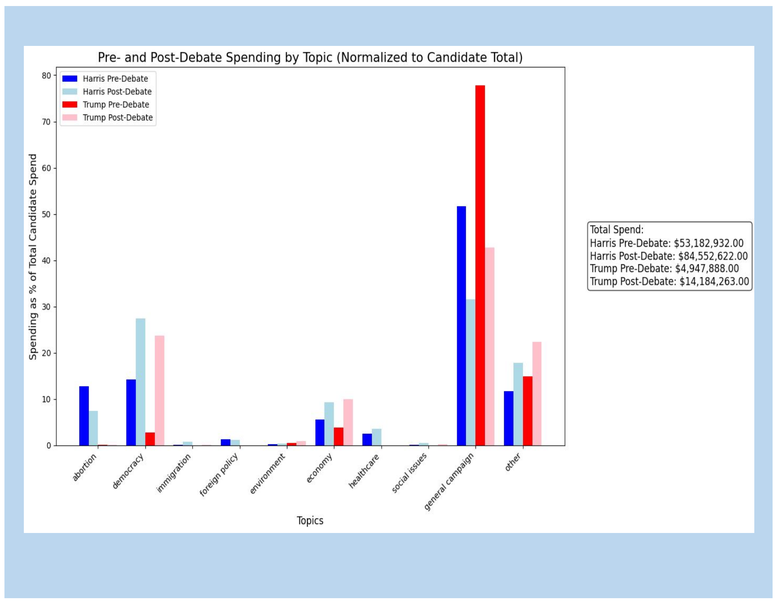


- We did not observe major changes between the two graphs.
- It is clearly demonstrated that the same topics were as popularly advertised for both candidates pre and post debate.
- While there are some minor differences, the candidates mostly seemed to target spending overall towards "general campaign" or donation advertisements and Harris had much higher spending than Trump

Dictionary Based Approach

- When results are scaled by percent of spend for each campaign, the results show that spending increased across the board after the debate (Figure 2).
- While we expected Harris to spend more on abortion rights messaging, her spending actually decreased
- Interestingly, while Harris' spending on democracy just about doubled after the debate, Trump's more than quadrupled to almost match hers
- The economy was the only issue where the spending flipped. Harris outspent Trump pre-debate, but he began outspending her after.

Figure 2. Dictionary Based Classification using REGEX



Discussion

- There are many factors that may lead a campaign to redirect spending, there were some significant changes made after the debate
- Harris outspent Trump by a very significant margin, but there were areas where they both directed most of their money
- While we expected certain outcomes (like Harris increasing spending on immigration), some of our expectations were not met (like Trump not spending on immigration)
- Additionally, we want to point out that a majority of the data was through the Harris/Biden campaign as they advertised more heavily on Facebook, which could have had a major effect on the data.
- The two methods also varied in the way that they filtered different values, categorized the ads by one topic in one approach and multiple topics in the other, amongst other small differences fit to each method/model. This may have contributed to the small discrepancies in our findings.
- To research this shift in spending further, we would want to train a formal model on previous campaign spending habits and ads to get a stronger set of keywords and topics that are typically covered during debate season
- Additionally, for the GPT model, the output was often variable due to the model returning a unique result/structure of the result every time (i.e. "Output: 'abortion'" vs "'abortion". We would like to adjust the model to make it more standardized and accurate.